

ECE 447

Lesson 22

Digital

Communication and Line Coding



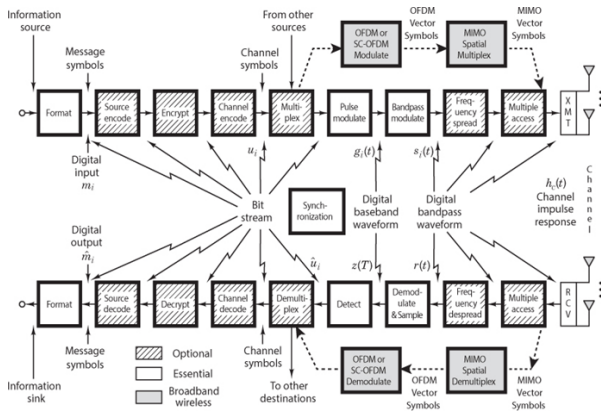
UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE AND ADMIN

- [Schedule](#)
- Admin
 - **HW3.** Grading...
 - **HW4.** Posted on course website. Due Lesson 23 (Tue, 13 Oct).
 - **Lab 5.** Due Lesson 24 (Thu, 15 Oct).

Message signal $m(t)$ → Source encoder → Baseband modulation (line coding) → Digital carrier modulation → Multiplexer → Channel → Regenerative repeater → ...

Below the Digital carrier modulation block, the binary sequence 1011010... is shown next to a waveform. Below the Multiplexer block, 'Other signals' are indicated by dashed arrows.

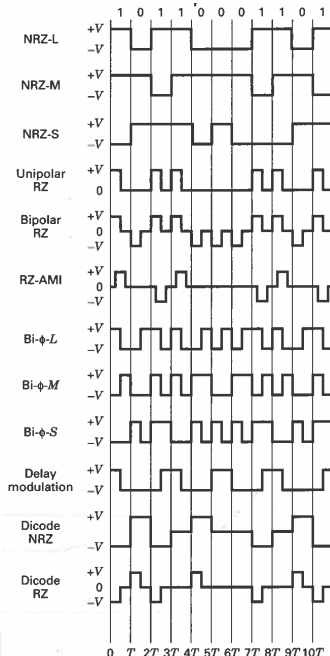
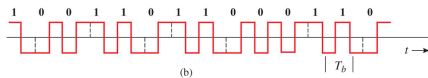
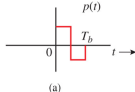
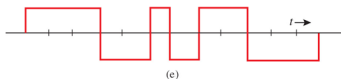
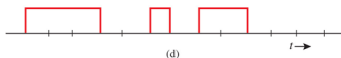
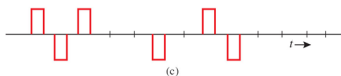
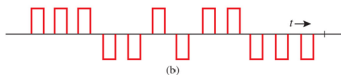
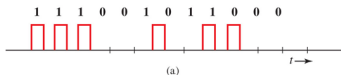


BASEBAND LINE CODING

Desired line code properties:

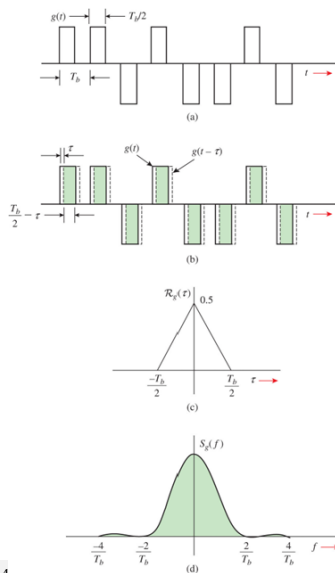
- Low bandwidth
- Power efficiency - low tx power for given bandwidth and error detection rate
- Error detection/correction capability
- No DC content
- Timing info (self-clocking)
- Transparency

TYPES OF LINE CODES



LINE CODE PSDs

Example 3.24 - Our good friend, Autocorrelation!



LINE CODE SPECTRA

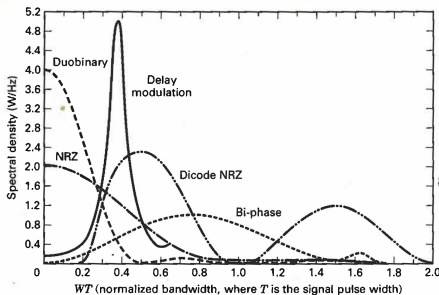
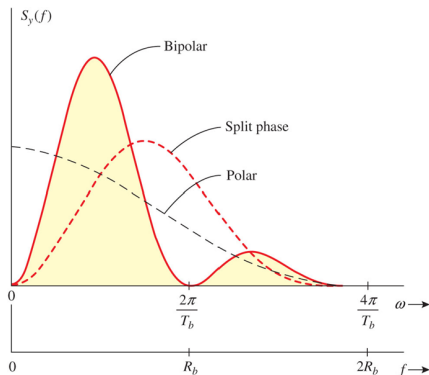


Figure 2.23 Spectral densities of various PCM waveforms.

LINE CODE TO BASEBAND WAVEFORM

